Laboratorium 9.2 — MARS

Statystyka w SI i analizie danych — ćwiczenia

Wojciech Maćkowiak, 412430

2026-05-05

Contents

Regresja — `Boston` 1

Klasyfikacja — `CarseatsH` 8

Podsumowanie 12

Pakiety: `earth`, `MASS`, `ISLR`, `plotmo`. Instalacja: `install.packages(c("earth", "MASS", "ISLR", "plotmo"))`.

```
library(earth)
library(MASS)
library(ISLR)
library(plotmo)
```

Regresja — `Boston`

Implementacja MARS jest w pakiecie `earth`.

```
head(Boston)
```

```
##      crim zn indus chas   nox    rm  age    dis rad tax ptratio  black lstat
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3 396.90  4.98
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8 396.90  9.14
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8 392.83  4.03
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7 394.63  2.94
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7 396.90  5.33
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7 394.12  5.21
##   medv
## 1 24.0
## 2 21.6
## 3 34.7
## 4 33.4
## 5 36.2
## 6 28.7
```

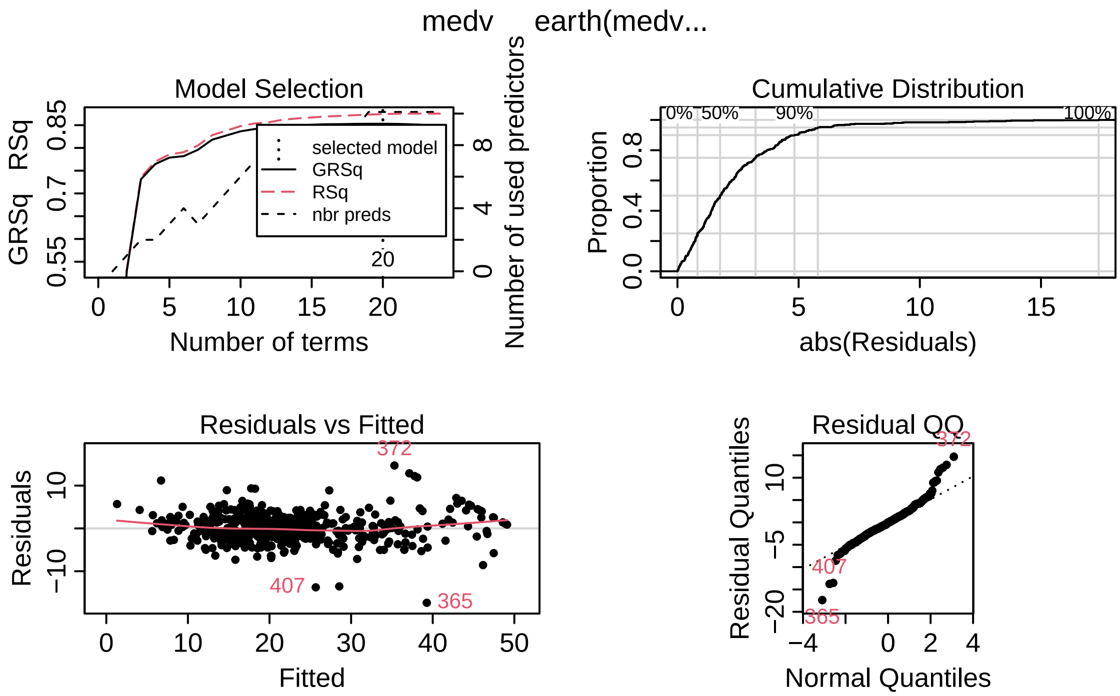
Model MARS bez interakcji wyższego rzędu (addytywny w sensie hinge functions):

```r
medv_mars <- earth(medv ~ ., data = Boston)
summary(medv_mars)
```

```
## Call: earth(formula=medv~., data=Boston)
##
##                 coefficients
## (Intercept)       -25.419924
## h(crim-4.42228)    -0.831429
## h(crim-11.0874)     0.741881
## h(4.93-indus)       0.637052
## h(nox-0.488)      -19.243817
## h(rm-6.431)         8.345864
## h(rm-7.929)        -7.544970
## h(dis-1.4165)      52.565495
## h(2.4358-dis)      55.189154
## h(dis-2.4358)     -53.319515
## h(3-rad)           -1.264920
## h(rad-3)            0.247856
## h(296-tax)          0.023509
## h(tax-296)         -0.006907
## h(14.7-ptratio)     1.363853
## h(ptratio-14.7)    -0.623227
## h(386.75-black)    -0.005366
## h(black-386.75)    -0.125594
## h(6.12-lstat)       2.311188
## h(lstat-6.12)      -0.450779
##
## Selected 20 of 24 terms, and 10 of 13 predictors
## Termination condition: Reached nk 27
## Importance: rm, lstat, ptratio, dis, nox, crim, tax, rad, black, indus, ...
## Number of terms at each degree of interaction: 1 19 (additive model)
## GCV 12.44743    RSS 5364.917    GRSq 0.853135    RSq 0.8744059
```

Wykresy diagnostyczne: selekcja modelu, ECDF rezyduów, rezydua vs fitted, QQ.

```r
plot(medv_mars)
```

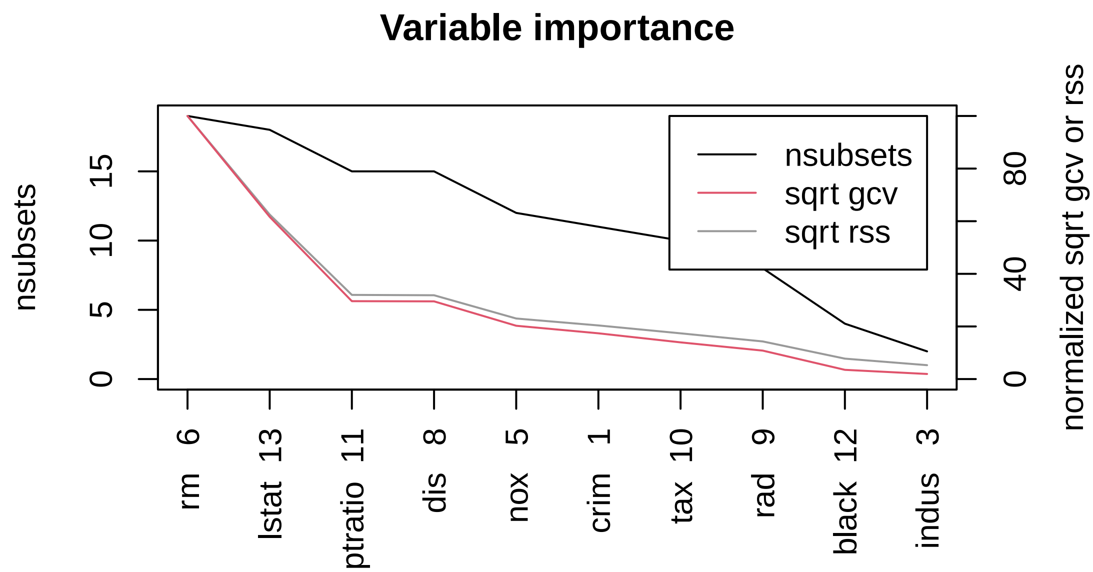


Ważność predyktorów (evimp):

```
medv_mars_evimp <- evimp(medv_mars)
medv_mars_evimp
```

```
##          nsubsets   gcv    rss
## rm             19 100.0  100.0
## lstat          18  61.6   62.5
## ptratio        15  29.6   32.0
## dis            15  29.5   31.9
## nox            12  20.3   23.0
## crim           11  17.4   20.4
## tax            10  14.0   17.4
## rad             8  10.8   14.3
## black           4   3.5    7.8
## indus           2   1.9    5.3
```

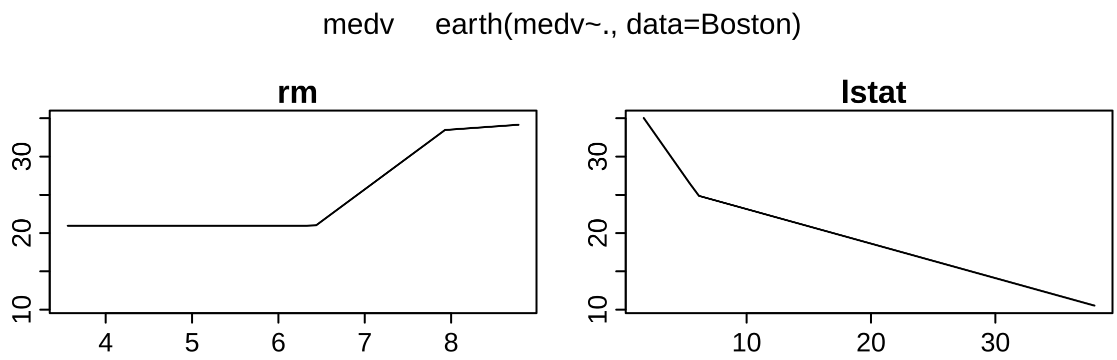
```
plot(medv_mars_evimp)
```



Częściowa zależność (`plotmo`, PDP):

```
plotmo(medv_mars, pmethod = "partdep", degree1 = c("rm", "lstat"))
```

```
## calculating partdep for rm
## calculating partdep for lstat
```



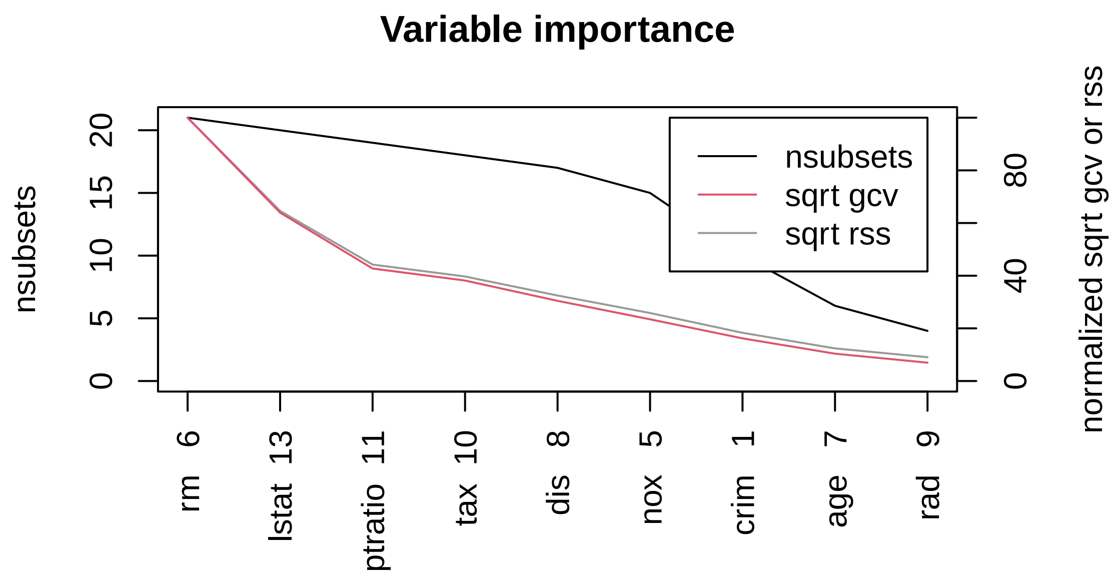
MARS z poziomem interakcji 2:

```
medv_mars_d2 <- earth(medv ~ ., data = Boston, degree = 2)
summary(medv_mars_d2)
```

```
## Call: earth(formula=medv~., data=Boston, degree=2)
##
##                                  coefficients
## (Intercept)                         26.162939
## h(4.42228-crim)                      0.512883
## h(crim-4.42228)                     -0.118817
## h(6.431-rm)                         27.371540
## h(rm-6.431)                          9.221378
## h(1.3567-dis)                      114.817027
## h(dis-1.3567)                       -0.824200
## h(lstat-6.12)                       -0.877722
## h(4.42228-crim) * h(age-20.8)       -0.011736
## h(4.42228-crim) * h(224-tax)         0.037938
## h(0.713-nox) * h(lstat-6.12)         2.906547
## h(nox-0.713) * h(lstat-6.12)         1.190195
## h(6.43-rm) * h(dis-1.3567)         -62.582248
## h(rm-6.43) * h(dis-1.3567)           0.472528
## h(6.431-rm) * h(dis-1.8209)         62.459177
## h(6.431-rm) * h(1.8209-dis)        -57.291677
## h(6.431-rm) * h(5-rad)              -1.929690
## h(rm-6.431) * h(ptratio-18.6)       -6.717680
## h(rm-6.431) * h(18.6-ptratio)        0.599069
## h(6.431-rm) * h(lstat-19.31)         0.323223
## h(307-tax) * h(6.12-lstat)           0.016902
## h(tax-307) * h(6.12-lstat)           0.017474
```

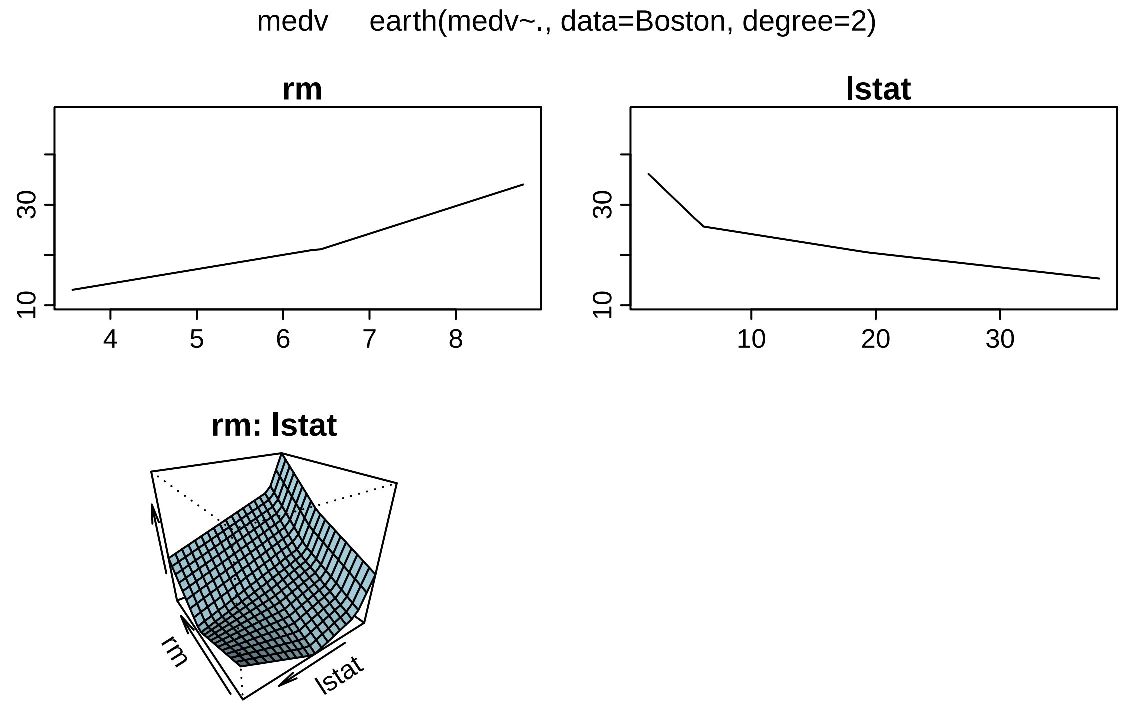
```
##
## Selected 22 of 27 terms, and 9 of 13 predictors
## Termination condition: Reached nk 27
## Importance: rm, lstat, ptratio, tax, dis, nox, crim, age, rad, zn-unused, ...
## Number of terms at each degree of interaction: 1 7 14
## GCV 9.045999    RSS 3660.524    GRSq 0.8932679    RSq 0.9143062
```

```r
evimp(medv_mars_d2) |> plot()
```



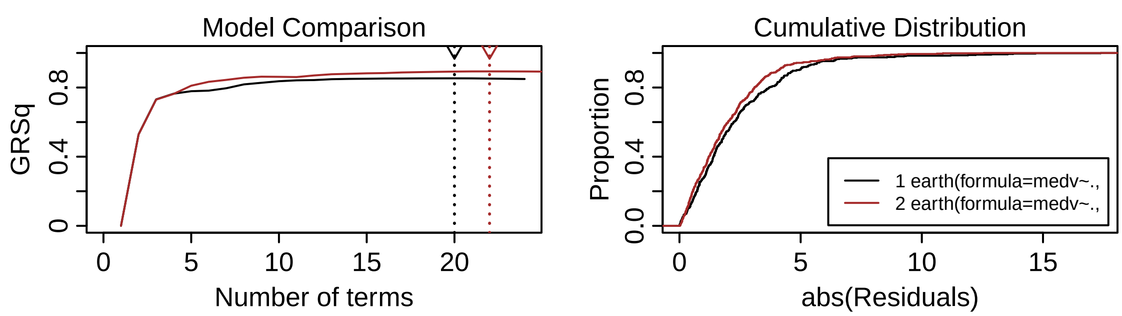

```r
plotmo(
  medv_mars_d2,
  pmethod = "partdep",
  degree1 = c("rm", "lstat"),
  degree2 = c("rm", "lstat")
)
```

```
## calculating partdep for rm
## calculating partdep for lstat
## calculating partdep for rm:lstat 01234567890
```



Porównanie modeli (RSS/GCV na zbiorze uczącym):

```
plot.earth.models(list(medv_mars, medv_mars_d2))
```



Oszacowanie MSE na zbiorze testowym — podział jak w laboratorium 8 (`set.seed(2025)`).

```r
set.seed(2025)
n_bo <- nrow(Boston)
train_bo <- sample(n_bo, 0.8 * n_bo)
test_bo <- setdiff(seq_len(n_bo), train_bo)

medv_mars_subset <- earth(medv ~ ., data = Boston, subset = train_bo)
medv_mars_subset_pred <- predict(medv_mars_subset, newdata = Boston[test_bo, ])
mean((medv_mars_subset_pred - Boston$medv[test_bo])^2)
```

```
## [1] 16.20058
```

```r
medv_mars_d2_subset <- earth(medv ~ ., data = Boston, subset = train_bo, degree = 2)
medv_mars_d2_subset_pred <- predict(medv_mars_d2_subset, newdata = Boston[test_bo, ])
mean((medv_mars_d2_subset_pred - Boston$medv[test_bo])^2)
```

```
## [1] 15.95257
```

Klasyfikacja — `CarseatsH`

```r
CarseatsH <-
  Carseats |>
  transform(High = factor(ifelse(Sales <= 8, "No", "Yes")))
head(CarseatsH)
```

```
##   Sales CompPrice Income Advertising Population Price ShelveLoc Age Education
## 1  9.50       138     73          11        276   120       Bad  42        17
## 2 11.22       111     48          16        260    83      Good  65        10
## 3 10.06       113     35          10        269    80    Medium  59        12
## 4  7.40       117    100           4        466    97    Medium  55        14
## 5  4.15       141     64           3        340   128       Bad  38        13
## 6 10.81       124    113          13        501    72       Bad  78        16
##   Urban  US High
## 1   Yes Yes  Yes
## 2   Yes Yes  Yes
## 3   Yes Yes  Yes
## 4   Yes Yes   No
## 5   Yes  No   No
## 6    No Yes  Yes
```

MARS z warstwą GLM (logit) — `glm = list(family = binomial)`:

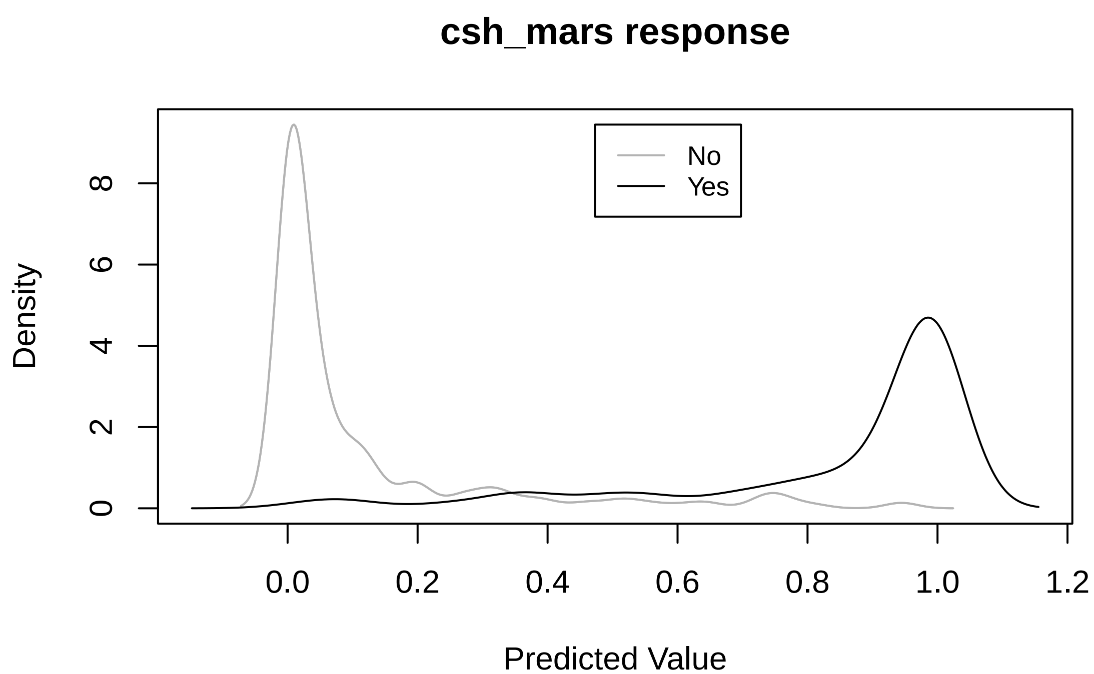
```r
csh_mars <- earth(High ~ . - Sales, data = CarseatsH, glm = list(family = binomial))
summary(csh_mars)
```

```
## Call: earth(formula=High~.-Sales, data=CarseatsH, glm=list(family=binomial))
##
## GLM coefficients
##                         Yes
## (Intercept)      -4.4303457
## ShelveLocGood     8.4408537
## ShelveLocMedium   3.6433533
## h(118-CompPrice) -0.1824056
## h(CompPrice-118)  0.1997402
```

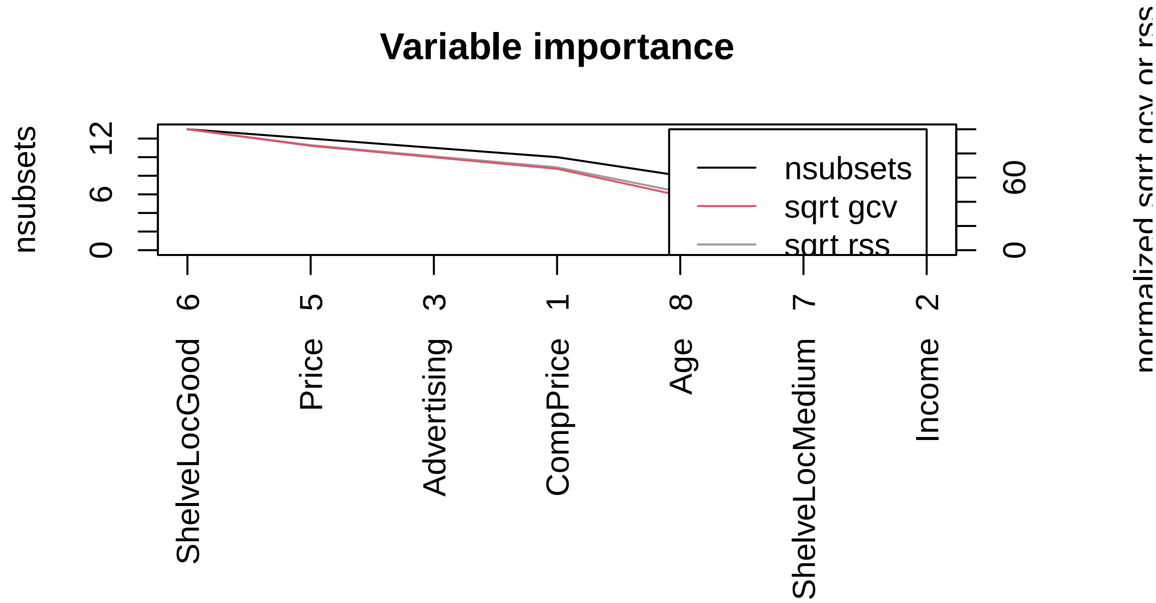
```
## h(CompPrice-153)  -0.5330426
## h(106-Income)     -0.0387567
## h(Advertising-3)   0.3385304
## h(Advertising-18) -1.2225360
## h(Advertising-20)  2.4847230
## h(110-Price)       0.1912332
## h(Price-110)      -0.1715017
## h(36-Age)          0.2376731
## h(Age-36)         -0.0621084
##
## GLM (family binomial, link logit):
##  nulldev  df       dev  df   devratio     AIC iters converged
##  541.487 399   169.184 386      0.688   197.2     7         1
##
## Earth selected 14 of 17 terms, and 7 of 11 predictors
## Termination condition: Reached nk 23
## Importance: ShelveLocGood, Price, Advertising, CompPrice, Age, ...
## Number of terms at each degree of interaction: 1 13 (additive model)
## Earth GCV 0.1084436    RSS 37.71914    GRSq 0.5539392    RSq 0.6101784
```

Diagnostyka (`plotd` z `earth`):

```r
plotd(csh_mars)
```




```r
evimp(csh_mars) |> plot()
```



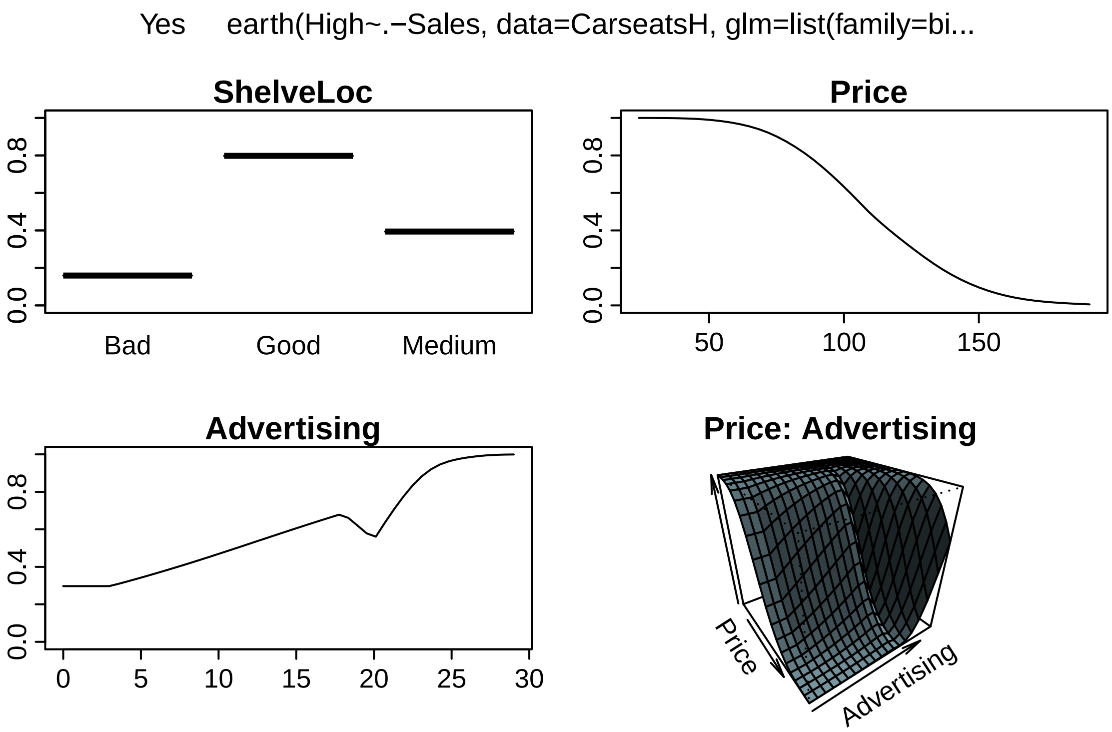

```r
plotmo(
  csh_mars,
  pmethod = "partdep",
  degree1 = c("ShelveLoc", "^Price", "Advertising"),
  degree2 = c("^Price", "Advertising")
)
```

```
## calculating partdep for ShelveLoc

## calculating partdep for Price

## calculating partdep for Advertising

## calculating partdep for Price:Advertising 01234567890
```



Błąd klasyfikacji na holdoucie (`set.seed(1)`):

```
set.seed(1)
n_cs <- nrow(CarseatsH)
train_cs <- sample(n_cs, 0.8 * n_cs)
test_cs <- setdiff(seq_len(n_cs), train_cs)

csh_mars_subset <- earth(High ~ . - Sales, data = CarseatsH, subset = train_cs, glm = list(family = bin
csh_mars_subset_prob <- predict(csh_mars_subset, newdata = CarseatsH[test_cs, ], type = "response")
csh_mars_subset_pred <- ifelse(csh_mars_subset_prob > 0.5, "Yes", "No")
mean(csh_mars_subset_pred != as.character(CarseatsH$High[test_cs]))
```

```
## [1] 0.125
```

```
csh_mars_d2_subset <- earth(
  High ~ . - Sales,
  data = CarseatsH,
  subset = train_cs,
  degree = 2,
  glm = list(family = binomial)
)
csh_mars_d2_subset_prob <- predict(csh_mars_d2_subset, newdata = CarseatsH[test_cs, ], type = "response
csh_mars_d2_subset_pred <- ifelse(csh_mars_d2_subset_prob > 0.5, "Yes", "No")
mean(csh_mars_d2_subset_pred != as.character(CarseatsH$High[test_cs]))
```

```
## [1] 0.1375
```

Podsumowanie

MARS (`earth`) dla regresji i klasyfikacji z GLM: modele addytywne i z `degree = 2`, ważność przez `evimp`, wizualizacje `plotmo` / `plotd`, MSE i błąd klasyfikacji na ujednoliconym podziale train/test jak w laboratorium 8.